DECISION · ORDERS PLATFORM

Which queue carries order events

Which message queue should carry order events from the start of next quarter?

RECOMMENDED	RUNNER-UP	DECIDE BY
SQS FIFO	**Keep polling**	**7 November 2026**
4.7 of 5, weighted	3.4 of 5	the platform lead

Context

Order events go from the orders API to billing, email and analytics through a table polled every five seconds. At 612 000 reads a day the polling costs more than the reads, and billing sees an order up to ten seconds late. Whatever replaces it must keep the events of one order in order.

How the options score

Criterion	Weight	Kafka	SQS FIFO	Keep polling
Keeps order per order	× 3	**5**	**5**	4
Little to run	× 3	2	**5**	4
Cost a month	× 2	2	**4**	1
Team knows it	× 1	3	4	**5**
Weighted score		3.1	**4.7**	3.4

Table 1: Each criterion is scored 1 (poor) to 5 (strong); the best in each row is in bold. The weighted score is out of 5.

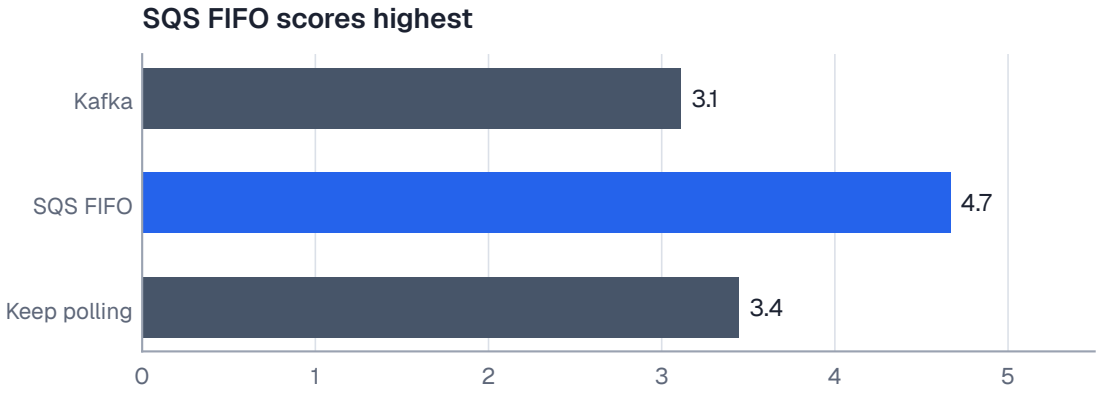


Figure 1: The weighted scores; the recommended option in colour.

The options

Kafka

A managed Kafka cluster, three brokers. (*Effort:* 6 weeks)

For

- Order kept per partition key
- Replays any day of events
- Analytics can read it directly

Against

- Brokers and partitions to size and watch
- The most expensive at our volume

SQS FIFO (recommended)

Amazon's FIFO queues, one message group per order. (*Effort:* 2 weeks)

For

- Nothing to run
- Order kept per message group
- Billing already reads SQS

Against

- No replay after a message is read
- 300 messages a second per group

Keep polling

The table as it is, polled every second instead of five. (*Effort:* 2 days)

For

- No new system

Against

- Five times the load on the database
- Still a second late

Recommendation

> **We recommend SQS FIFO**
> It keeps each order's events in order with nothing to run, at a third of Kafka's cost, and billing already reads SQS.

What would change our mind

- Analytics needs to replay events older than four days.
- Order events pass 300 a second for a single order (bulk imports).

Risks

- A consumer that fails mid-group holds that order's later events until it recovers.

Next steps

1. Agree the message group key (the order ID) with billing.
2. Move email first, then billing, then analytics.
3. Retire the polled table once all three read the queue.